AI & SOCIETY COURSE

Prof. Mor Naaman

Unit 0

WELCOME TO THE COURSE!

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Today

- Who am I?
- Why am I doing this?
- What are we going to do here?
- How is this course going to work?

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First things first

Who am I?

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My background

- Professor of Information Science at Cornell University (Cornell Tech campus, New York City)
- PhD in Computer Science (Stanford)
- Startup experience
- Big company experience: Yahoo, Google
- Yes, I played (professional) basketball.

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Information Science vs. Computer Science

Computer Science

Builds the technology: algorithms, systems, models.

Information Science

Studies the technology in the world: how people use it, and what is it's impact on society.

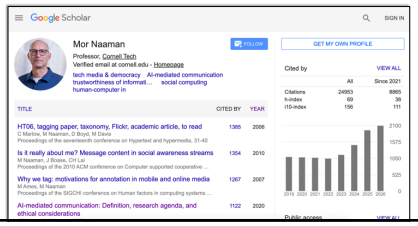


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My research

The impact of technology (most recently AI) on our communication and information environment.



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Cornell Tech: the campus





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Cornell Tech: the campus and programs

- Cornell University's graduate campus in New York City, on Roosevelt Island
- Master's and PhD programs in technology (CS/ECE/DS/…), design, business, law, and information science
- Built around building things: students work with startups and companies from day one



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Cornell University: main campus

- Cornell University's main campus in upstate New York
- Undergraduates, Master's and PhD programs





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Motivation

Why am I doing this?

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Almost 10 years of studying AI's impact

- I have been studying the impact of AI on people and society for almost a decade
- When we started, this technology was not even there: no ChatGPT, no AI in anyone's hands
- Everyone is asking these questions now…
- … But not everyone have the same background knowledge

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Why this course

- Helping students and young professionals across the world know more about AI and its impact
- Learn from these students (you!) about how they think about AI, and what they think is important

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The plan

What are we going to do here?

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Our focus: Generative AI and LLMs

- This is not a course about the impact of AI in general, or about machine learning in general
- We will focus on Generative AI and large language models (LLMs)
- We will explain what "Generative AI" actually means, and how LLMs work
- ChatGPT and similar chatbots are our main example

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Beyond the chatbot

- Access to models through an API
- Image generators
- Voice
- Meeting summarizers
- Video
- Agents…

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Learning goals

- Understand how AI works
- Understand what its limitations are
- Understand its consequences for people and for society
- Think about the future of AI in your own country

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> *"To understand how LLMs will change the world, we will need to understand what they are capable of, and what they are not."*
>
> — Carl Bergstrom & Jevin West, *Calling Bullshit*

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What this course is NOT

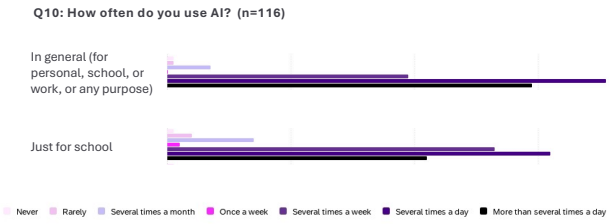
- **NOT "how to use AI"**
- **NOT "which AI tools to use"**

But: you will build an understanding of these systems that will help you make better use of AI — and that will hold up as the tools keep changing.

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Some survey results



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Who is in the room?

- Techies and non-techies are welcome
- You do not need to know anything about AI or computer science to follow this course
- In the first part, people with a CS/ML education may already know some of the material

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Critical, but not anti-AI

- I take a very critical view of the challenges and risks of AI
- That does not make this an anti-AI course
- There are plenty of promoters of AI online, and the companies building it
- The critical view is the one you are less likely to hear, and you need it as users, as developers, and as future policy makers

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The six units

What you will learn

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Unit 1 — AI Foundations

How LLMs and chatbots actually work, the data and compute behind them, and where they are biased and where they fail.

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Unit 2 — AI and Individuals

How using AI changes the way you think, learn and create; how you come to trust and rely on it, and how it can be designed to persuade you.

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Unit 3 — AI, Human Communication, and Information

How AI reshapes the way people communicate, connect and evaluate each other; and what it does to culture, news, and what we believe is true.

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Unit 4 — Challenges for AI Implementation

The harms AI can cause and how they are prevented, who owns the data and the outputs, and whether the economics of AI hold up.

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Unit 5 — AI's Worldwide Impact

How AI is disrupting education, work, power and democracy, and what are the "hidden" costs of AI for people and for the planet.

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Unit 6 — The Way Forward with AI

What it would take to make positive use of AI in your own community and your own country.

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Ground rules

How is this course going to work?

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The same course I teach at Cornell

- This is the same material I would teach Cornell undergraduates
- And the same style of teaching
- Only over a small number of days instead of a semester...

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Interactive course!

- There are lectures, but also discussion, exercises, and participation
- Interrupt me at any time
- Disagree with me, or correct me
- Tell me something I do not know!

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English

- It is okay if your English is not perfect
- My English is not perfect either (and I have an accent)
- Stop me at any time if you don't get something: I want you to understand

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Long days

- Many hours a day – will be hard for you and me
- We'll take breaks
- We'll do some exercises
- But we will also get tired…

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This is the second time!

- Things will break. Things may not work. Something might not be clear.
- We will work it out together

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I don't know what you don't know.

So stop me and ask.

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Laptops and phones

- You will be asked to use a laptop/computer/phone in class for the exercises
- I ask that you do NOT use them for anything else, and keep them off when you are not doing an exercise
- Paper and pen are welcome for notes; I will make the slides available

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AI policy

- You will be asked to use AI services some of the time
- Any other time, AI use is not allowed.
- If you do not have sufficient AI access, I can arrange access to some services for the duration of the course; come to see me during break.

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AI disclosure

- I created all of the content in this course myself
- I often used Claude to create the actual slides, and reviewed the results
- I take full responsibility for the content of each and every slide, and its accuracy. All errors (if any) are mine.

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